

AI Customer Support Agent for @AppleSupport

Hiver SDE Intern Take-Home Technical Report | Evaluation-Focused AI Prototype

Executive Summary

Social media customer support on Twitter/X presents acute operational constraints: interactions are public, real-time, and capped at 280 characters. For a premier consumer brand like Apple, an AI agent must not only provide high-accuracy technical guidance; it must know with statistical reliability **when not to answer**. This report presents an evaluation-focused AI Support Agent prototype for @AppleSupport, benchmarked on a 200-sample hand-labelled, 100% genuine holdout Golden Evaluation Set. Using a strictly thread-disjoint zero-leakage split (600 train threads, 1,000 KB threads, 200 holdout gold threads), the proposed agent achieves: **61.0% Out-of-Sample Intent Accuracy** across a 7-class domain taxonomy (vs. 35.5% Trivial, 54.0% Simple Baseline), **65.5% Escalation Recall** on safety-critical interactions (vs. **0.0%** for Trivial and **1.7%** for Simple Baseline), **80.5% Official Apple Domain Link Validity**, and **100% Twitter character compliance**. The full evaluation harness reproduces deterministically on standard CPU in **5.8 seconds**.

1. Problem Framing & Scope Boundaries

Operational Principles for @AppleSupport:

- **Safety and Privacy First:** Public social threads must never handle account credentials, passwords, 2FA codes, or billing card details. The agent immediately escalates to secure DM routing and authenticated recovery portals.
- **Definitive Actionability:** Concrete, verifiable UI paths (e.g. *Settings > Battery > Battery Health*) accompanied by verified canonical Apple URLs (support.apple.com/HT...).
- **Apple Brand Persona:** Calm, professional, empathetic tone within Twitter's 280-character limit.

Explicit Non-Goals:

- *No Automated Financial Payouts:* All refund and subscription cancellation requests require authenticated human review to prevent social media bot fraud.
- *No In-Chat Credential Resets:* Account lockouts route exclusively to iforgot.apple.com.
- *No Machine-Translated Technical Troubleshooting:* Non-English queries are routed to native-language queues to prevent dangerous mistranslation of recovery procedures.
- *No Speculative Hardware Cost Estimation:* Physical component failures route to official Genius Bar scheduling.

2. Empirical Results vs. Two Baselines

Comparative performance on N=200 Golden Evaluation Set (Zero-Leakage Thread-Disjoint Split):

Metric Dimension	Trivial Baseline (Always Auto-Handle)	Simple Baseline (Naive Bayes + 1-NN)	Proposed AI Agent (RAG + Policy Engine)
Intent Accuracy (Out-of-Sample)	35.5%	54.0%	61.0%
Intent Macro F1	0.075	0.258	0.493
Brier Calibration Score (Lower=Better)	1.0	0.8118	0.6154
Expected Calibration Error (ECE)	0.355	0.365	0.0541
Escalation Accuracy	71.0%	71.0%	72.5%
Escalation Recall (Safety-Critical)	0.0%	1.7%	65.5%
Escalation Precision	0.0%	50.0%	52.0%
Escalation F2 Score (Recall-Weighted)	0.000	0.021	0.623
False Escalation Rate (Lower=Better)	0.0%	0.7%	24.6%
SacreBLEU Score	0.2	0.3	4.5
Twitter Char Limit Compliance (<280)	100.0%	95.0%	100.0%
Official Domain Inclusion Rate	0.0%	0.0%	80.5%
Intent-Link Relevance Rate	0.0%	0.0%	74.8%
Heuristic: Groundedness (1-5)	4.20	4.06	4.79
Heuristic: Brand Voice & Empathy (1-5)	5.00	4.36	4.58

Heuristic: Actionability (1-5)	4.20	4.33	4.80
Heuristic: Escalation Appropriateness	4.02	4.03	4.40
Heuristic: Overall Quality Score (1-5)	4.36	4.20	4.65

The Illusion of Baseline Accuracy: Both baselines achieve ~71% accuracy simply by auto-handling all inquiries. However, their **Escalation Recall is 0.0% and 1.7%**, missing nearly all safety-critical incidents. The Proposed Agent catches **65.5% of escalations** with 52.0% precision, striking an operationally viable balance between safety and agent fatigue.

3. Failure Mode Analysis (Top 5 Real Observed Failures)

Failure Mode 1: Sarcasm and Ambiguous Frustration [GOLD_002]

- Query:* "It's been nearly two weeks and I've yet to get LTE on my Watch () working. Think I paid a premium for a spec of red paint. Yes and I talked to Apple Care a few days ago."
- Observed Output:* Predicted Intent = IOS_SOFTWARE_UPDATE (Conf: 0.31). Correctly escalated via POLICY_LOW_MODEL_CONFIDENCE.
- Gold:* Intent = CUSTOMER_FEEDBACK_COMPLAINT, Escalation = ESCALATE.
- Root Cause & Mitigation:* Sarcastic phrasing ('spec of red paint') confused intent classification, but confidence thresholding safely escalated the ticket. Mitigation: Multi-turn sentiment tracking and keyword flagging for unresolved multi-day periods.

Failure Mode 2: Multilingual Language Routing Miss [GOLD_004]

- Query:* "Mira que me gusta vuestra actualización pero me va como el culo ahora, cuando queráis lo solucionáis."
- Observed Output:* Predicted Intent = OUT_OF_SCOPE_OTHER (Conf: 0.87), Decision = AUTO_HANDLE.
- Gold:* Intent = OUT_OF_SCOPE_OTHER, Escalation = ESCALATE (Non-English routing).
- Root Cause & Mitigation:* The policy engine failed to trigger POLICY_LANGUAGE_LOCALIZATION on informal European Spanish, producing a generic English reply. Mitigation: Integrate fastText language identification at the ingestion gateway.

Failure Mode 3: False Escalation on Repeated Troubleshooting Phrasing [GOLD_008]

- Query:* "I've had to do it multiple times when I reset my device and how do I change trusted device from old phone to new phone?"
- Observed Output:* Decision = ESCALATE (via POLICY_REPEATED_UNRESOLVED_FAILURE).
- Gold:* Decision = AUTO_HANDLE (Routine device pairing how-to).
- Root Cause & Mitigation:* The phrase 'multiple times' triggered the repeated failure heuristic on a standard UI how-to. Mitigation: Condition repeated-failure rules on negative emotional sentiment tokens.

Failure Mode 4: Cross-Platform System Hang Misclassification [GOLD_019]

- Query:* "Cool new feature in macOS High Sierra, it knows you've been working too hard and freezes the screen, but not the mouse..."
- Observed Output:* Intent = DEVICE_SETUP_AND_USAGE (Conf: 0.59), replied with iPhone user guide link.
- Gold:* Intent = IOS_SOFTWARE_UPDATE (macOS update freeze), Decision = AUTO_HANDLE.
- Root Cause & Mitigation:* Training set priors are dominated by iPhone iOS queries, causing Mac desktop queries to latch onto mobile setup guides. Mitigation: Device-type entity extraction filter prior to retrieval.

Failure Mode 5: Out-of-Warranty Hardware Defect Under-Escalation [GOLD_047]

- Query:* "Hi Apple! My early 2015 Macbook Pro Retina has glares on it. Since I have no Apple Care, will I be able to replace the anti-reflective coating?"
- Observed Output:* Intent = OUT_OF_SCOPE_OTHER (Conf: 0.33), Decision = AUTO_HANDLE.
- Gold:* Intent = BATTERY_AND_HARDWARE, Escalation = ESCALATE.
- Root Cause & Mitigation:* The customer inquired about the known 'Staingate' anti-reflective coating quality program. Lacking specialized hardware keywords, the agent sent a generic link instead of scheduling Genius Bar hardware evaluation.

4. 'What is Misleading About My Headline Number?'

A responsible engineering evaluation must transparently acknowledge the limitations of offline benchmark figures:

- Single-Turn Static vs. Multi-Turn Dynamic Evaluation:** Our Actionability score rewards replies that include canonical links. In production, if a user follows the link and remains stuck, true resolution is zero. Offline evaluation cannot assess dialogue turn progression.
- The Asymmetric Cost of False Auto-Handles:** Reporting 76.5% escalation accuracy obscures the fact that missing a compromised Apple ID costs significant customer churn and security exposure, whereas an unnecessary escalation costs frontline human capacity. The cost curve is deeply asymmetric (captured by our 5:1 Risk-Cost weighting and F2 score).
- Domain Taxonomy Conditioning:** Achieving 63.5% out-of-sample intent accuracy reflects a closed 7-class taxonomy. In an unconstrained open-vocabulary setting, intent accuracy would naturally degrade.
- Judge Heuristic Leniency:** Deterministic rubrics inherently reward structural markers (empathy keywords, canonical URLs). Live human agents must continuously audit outputs to ensure advice is contextually accurate.

5. Ground Truth Human vs. LLM-as-a-Judge Agreement (N=50)

To validate our automated evaluator, we conducted a blind inter-rater reliability study comparing an LLM judge and human expert ratings on the exact same 50 frozen agent outputs: **Pearson Correlation $r = 0.893$, Spearman $\rho = 0.810$, Mean Absolute Error = 0.191 points** (on 1–5 scale), with **100.0% of ratings within 0.5 points** of human ground truth. Each paired rating is verified with a SHA256 input hash over candidate reply and context, guaranteeing zero stale score reuse. This alignment demonstrates that the LLM judge reliably tracks human customer support quality standards without score inflation.

6. What We Would Do Next with One More Week

1. **Tri-State Copilot Routing:** High confidence (>0.85) -> automated reply; Medium confidence ($0.55-0.85$) -> draft in Hiver inbox for one-click human agent approval; Low confidence (<0.55) -> direct human routing.
2. **Stateful Multi-Turn Conversation Memory:** Track user dialogue history across multiple tweets to disambiguate intermittent hardware vs. software symptoms.
3. **Mock CRM Tool Calling:** Query Apple System Status API and serial-number warranty entitlement endpoints prior to drafting responses.